CURRICULUM VITAE

Isaac W. Sluder

Department of Mechanical Engineering, The University of Akron
Auburn Science & Engineering Center, Akron, OH 44325-3903
Tel: (330) 301-6790 | Email: iws4@uakron.edu

EDUCATION

The University of Akron, Akron, Ohio (8/2024-7/2028)
Doctor of Philosophy in Engineering, Mechanical Engineering
Advisor: Dr. K.T. Tan

The University of Akron, Akron, Ohio (8/2019-5/2024)
Bachelor of Science (Magna Cum Laude), Mechanical Engineering
Minor in Applied Mathematics
GPA: 3.771/4.000

RESEARCH EXPERIENCE & EMPLOYMENT

The University of Akron, Akron, OH
Graduate Research Assistant, Department of Mechanical Engineering (5/2024-Now)

- Conducted experiments to analyze the environmental effects of water and temperature on 3D printed composite materials.
- Architected a Large Language Model (LLM) workflow to aggregate environmental degradation data, streamlining the creation of a comprehensive database for predictive material modeling.
- Authored an experimental paper on hygrothermal effects in additively manufactured polyamide composites (accepted at *Composites Part B: Engineering*, 2026).

The University of Akron, Akron, OH
Graduate Teaching Assistant, Department of Mechanical Engineering (8/2024-Now)

- Led an undergraduate senior lab on mechanical pump systems, guiding students in hands-on experimentation and data analysis.
- Instructed a Concepts of Design lab, helping students develop a strong foundation in material selection and mechanical design principles.
- Designed and implemented a freshman Arduino & coding project to design a Bluetooth remote controlled vehicle, introducing students to embedded systems and hands-on programming.

Air Force Research Lab, Wright-Patterson AFB, OH
Research Intern, Polymer Matrix Composites Branch, Foundational Technologies Directorate (5/2026-8/2026)

- Developed an end-to-end pipeline using Gemini 3.5 Flash to extract and normalize chemical and materials data from scientific literature into a curated knowledge graph.
- Built a literature-grounded query system that generated interactive reports, data visualizations, and chemical structure diagrams with traceable source evidence.

- Reduced the estimated processing of a 1,500-paper corpus from 18 weeks of full-time chemist labor and approximately $35,000 in salary costs to 20 hours and approximately $140 in compute and API costs.

M. K. Morse, Canton, OH
Mechanical Engineering Co-Op, R&D (1/2022-5/2024)

- Spearheaded the development of a cutting-edge carbide tip orientation sorter utilizing Machine Learning vision technology, enhancing precision and efficiency in production processes.
- Engineered a sophisticated sensor system tailored for bandsaw back edge profiling, yielding significant cost reductions and optimizing operational performance.
- Led proprietary research initiatives at the intersection of cutting-edge technologies and artificial intelligence, contributing to the company's strategic innovation roadmap.
- Innovated and executed designs for cutting tools and associated testing apparatus to support both research endeavors and product development initiatives, ensuring alignment with industry standards and market demands.

The University of Akron, Akron, OH
Undergraduate Research Student, R&D (9/2023-5/2024)

- Executed comprehensive mechanical testing protocols on 3D printed carbon fiber reinforced polymers (CFRP), contributing to the advancement of materials science research.
- Initiated the mechanical recycling of CFRPs to evaluate the effects of reprocessing on material properties and sustainability.
- Facilitated weekly progress review meetings and presented findings and developments into concise presentations.
- Disseminated research results to the academic community by contributing to a manuscript and presenting at a local student conference.

The University of Akron, Akron, OH
Scanning Electron Microscope Manager, Department of Mechanical Engineering (2/2023-8/2023)

- Orchestrated comprehensive training sessions to acquaint new users with SEM microscope functionalities, ensuring proficiency and adherence to best practices.
- Spearheaded equipment management efforts, diligently addressing downtime and swiftly resolving technical issues to sustain uninterrupted workflow.
- Applied advanced testing methodologies to evaluate material properties essential for cutting-edge AI research, leveraging CNC scratch testing techniques to deliver precise and insightful data analysis.

The University of Akron, Akron, OH
Research Assistant, Department of Mechanical Engineering (6/2021-12/2021)

- Executed rigorous testing protocols on oils and greases, meticulously adhering to Standard Operating Procedures (SOPs) to maintain consistency and reliability.
- Employed data analysis techniques to extract valuable insights, subsequently optimizing processing times for enhanced efficiency and productivity.

Yellow Creek Casting Co., Wellsville, OH
Foundry Laborer (12/2018-5/2019)

- Specialized in mold-making within a foundry setting, proficiently utilizing green sand techniques and pouring molten iron into molds to craft intricate components.
- Gained foundational insights into materials science dynamics by actively engaging with molten iron, fostering a deep understanding of its behavior throughout the casting process.
- Exhibited meticulous precision and a keen eye for detail throughout mold-making operations, bolstering the production of high-quality cast iron components.
- Previously employed as a janitor from January 2015 to November 2018, responsible for maintaining cleanliness in office and lunchroom facilities, demonstrating a strong work ethic and reliability.

RESEARCH INTERESTS

Materials Informatics, Knowledge Graphs, Literature Data Extraction, LLM Workflows, Impact Dynamics, Environmental Polymeric Degradation, Materials Science, Composites, Biomimicry, Large Language Models, Data-Science, Machine Learning, Statistical and Mathematical Analysis.

HONORS & AWARDS

- **EDA Sustainable Polymers Tech Hub Workforce Initiative,** Sustainable Environment (WISE) grant (04/2026)
- **F3 RubberCity**, F3 HOF Leadership Award (04/2026)
- **Ohio Space Grant Consortium,** Summer Internship (06/2025 – 08/2025)
- **Ohio Space Grant Consortium,** Master's Fellowship (01/2025 – 05/2025, 08/2025 – 12/2025)
- **National Science Foundation,** National Research Traineeship (08/2025 – Now)
- **American Society for Composites,** 2025 Student Travel Award (10/2025)
- **American Society for Composites,** 2024 Student Travel Award (10/2024)
- **The University of Akron,** Mechanical Engineering Graduate Scholarship (08/2024 – Now)
- **The University of Akron,** Mechanical Engineering Under-Graduate Scholarship (2020)
- **The University of Akron**, Dean's List (08/2019 – 05/2020), (8/2021 – 5/2023)
- **The University of Akron**, President's List (08/2023)

STUDENT AWARDS & RECOGNITION

- Elementary Team (Mentor: Jensen Wu, Kyler Bedzyk, Olivia Li, Rishi Shah), Presidential AI Challenge, Ohio Track I State Winner (Project: *GooseBot: An AI-Powered Solution for Cleaner and Safer Community Parks*), 2026

PEER REVIEWED JOURNAL PUBLICATIONS

1. **Sluder, I. W.**, Frankowski, A. P., Tan, K. T., "Hygrothermal Degradation of Tensile and Impact Properties of 3D-Printed Chopped Carbon Fiber Composites", *Composites Part B: Engineering*, 327, 114107 (2026), 10.1016/j.compositesb.2026.114107

Manuscripts in Preparation/Submission

1. Frankowski, A. P., **Sluder, I. W.**, Tan, K. T., "Mechanical Recycling and Additive Re-manufacturing of Nylon-based Short Carbon Fiber Composites", *Submitted to Composites Part B.*

CONFERENCE PUBLICATIONS & ORAL PRESENTATIONS

1. Sluder, I. W., Cheng, E., Tan, K. T., Large Language Model–Driven Automation of Hygrothermal Degradation Data Extraction for Thermoplastic Composites, **American Society for Composites 41st Technical Conference**, Sep 28-30, 2026, Madison, Wisconsin, USA, **Upcoming Oral Presentation**
2. Sluder, I. W., Tan, K. T., Predicting Mechanical Degradation in 3D-Printed Nylon-Based CFRP from Environmental Factors, **American Society for Composites 40th Technical Conference**, Oct 5-8, 2025, Dayton, Ohio, USA
3. Sluder, I. W., Frankowski, A. P., Tan, K. T., Hygroscopicity of 3D Printed Carbon Fiber Reinforced Polymeric Composites, **American Society for Composites 39th Technical Conference***,* Oct 21-24, 2024, San Diego, California, USA
4. Frankowski, A. P., Sluder, I. W., Tan, K. T., Mechanical Recycling of 3D Printed Carbon Fiber Reinforced Composites for Increased Sustainability Through a Reduction of Waste Material, **American Society for Composites 39th Technical Conference**, Oct 21-24, 2024, San Diego, California, USA
5. Ramachandran, M. K., Sumaiya, S. A., Golvaskar, M., Sluder, I. W., Rakurty, C. S., Rangasamy, N., Emuakpor, O. S., Kannan, M., Improving the Fatigue Life of an Additively Manufactured Stainless Steel Specimen Using a Secondary Grinding Process, **ASME Turbo Expo 2024 Conference**, June 25–27, 2024, London, England, United Kingdom
6. Sluder, I. W., Curtis, P. M., Tan, K. T., Curved Fiber Patterns in 3D-Printed Carbon Fiber Composites: A Flexural Study, **AIAA Region III 2024 Conference**, April 5-6, 2024, Akron, Ohio, USA
7. Frankowski, A. P., Sluder, I. W., Tan, K. T., Characterizing the IZOD Impact Strength of a 3D Printed Reinforced Carbon Fiber Composite Material with Varying Layup Orientations, **AIAA Region III 2024 Conference**, April 5-6, 2024, Akron, Ohio, USA

VOLUNTEER & OUTREACH ACTIVITIES

- **The Chapel**, Worship Team Leader, and Volunteer (05/2025-Now)
- **F3 Akron**, Site Little-Q (07/2026–Now)
- **F3 Akron**, Volunteer Leader (Q) (05/2025–Now)
- **Presidential AI Challenge – Hudson City Schools**, Coach (9/2025-1/2026)
- **The Chapel**, Campus Focus, Worship Team Leader, and Volunteer (08/2019-05/2025)
- **The Chapel**, Campus Focus, Small Group Leader (08/2020-05/2024)
- **Science Olympiad,** Grader (03/2024)
- **The Chapel**, Trunk or Treat Event (10/2022, 10/2023, 10/2025)
- **Haven of Rest Ministries,** Volunteer (2019)

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS

1. American Society for Composites (ASC) 2024–Now
2. American Institute for Aeronautics and Astronautics (AIAA) 2024–2025

TECHNICAL SKILLS

Computer Software: Google Cloud, Vertex AI, OpenRouter, Langflow, Python (Jupyter Lab, VS Code), MATLAB, C++ (Arduno, Esp32), R (R-Studio), PLC (Allen Bradley), SQL (Influxdb, SQLite), Julia, CLI (Powershell, Zsh, Bash), Grafana, Microsoft Applications, OriginPro, MiniTab, SolidWorks, Fusion 360
Experimental Techniques: μCT microscope, Additive manufacturing machines, Environmental chambers, Dynamometer (Kistler), Instron 5582 Universal Testing Machine, Instron CEAST 9350 impact testing, SEM – Tescan Lyra 3 XMU with EDAX, Keyence VHX-7000, Penetrometer, Zygo Microscope, MPR – PCS Instruments, High-Frequency Reciprocating Rig, Ultra-High Pressure Viscometer, Rheometer

PROFESSIONAL REFERENCES

- **Dr. Kwek-Tze "K.T." Tan**
  Professor
  Department of Mechanical Engineering
  The University of Akron
  Auburn Science & Engineering Center, Rm N113, Akron, OH 44325-3903
  Tel: 330-972-7184, Fax: 330-972-6027
  E-mail: ktan@uakron.edu

- **Dr. Vikas Varshney**
  Senior Research Scientist
  Polymer Matrix Composites Branch, Foundational Technologies Directorate
  The University of Akron
  Air Force Research Laboratory, Wright-Patterson Air Force Base, OH
  Tel: 385-636-1051
  E-mail: vikas.varshney.2@afrl.af.mil

- **Dr. Sekhar Rakurty**
  R&D Manager
  The M. K. Morse Company
  1101 Eleventh Street SE, Canton, OH 44707
  Tel: 330-453-8187
  E-mail: rakurtys@mkmorse.com

- **Mr. Mark Kovscek**
  Founder and CEO
  Conservation Labs
  Materials and Structures Division, Cleveland, OH 44135
  Tel: 724-208-1962
  E-mail: mark@conservationlabs.com

- **Dr. Manigandan Kannan**
  Assistant Professor
  Department of Mechanical Engineering
  The University of Akron
  Auburn Science & Engineering Center, Rm 422B, Akron, OH 44325-3903
  Tel: 330-972-8672
  E-mail: mk77@uakron.edu